```
from google.colab import files
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Upload image
uploaded = files.upload()
filename = list(uploaded.keys())[0]

# Read image
image = cv2.imread(filename)
print("Image uploaded successfully:", filename)

# Get image size
height, width = image.shape[:2]
print("Width:", width)
print("Height:", height)

# Select 3 source points
source_points = np.float32([
    [50, 50],
    [200, 50],
    [50, 200]
])

# Set 3 destination points
destination_points = np.float32([
    [20, 80],
    [220, 50],
    [80, 220]
])

# Calculate affine transformation matrix
matrix = cv2.getAffineTransform(
    source_points, destination_points
)

print("Affine Matrix:")
print(matrix)

# Apply affine transformation
affine_image = cv2.warpAffine(
    image, matrix, (width, height)
)

# Display original and transformed images
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(affine_image, cv2.COLOR_BGR2RGB))
plt.title("Affine Transformed Image")
plt.axis("off")

plt.show()
```

Choose Files Screenshot… 075823.png

Screenshot 2026-10-10 075823.png(image/png) - 93416 bytes, last modified: 10/10/2026 - 100% done

```
Saving Screenshot 2026-10-10 075823.png to Screenshot 2026-10-10 075823 (2).png
Image uploaded successfully: Screenshot 2026-10-10 075823 (2).png
Width: 299
Height: 196
Affine Matrix:
[[  1.33333333   0.4        -66.66666667]
 [ -0.2          0.93333333  43.33333333]]
```

Original Image



Affine Transformed Image



Getting started

The document you are reading is not a static web page, but an interactive environment called a **Colab notebook** that lets you write and execute code.

For example, here is a **code cell** with a short Python script that computes a value, stores it in a variable, and prints the result:

```
seconds_in_a_day = 24 * 60 * 60
seconds_in_a_day
```

```
86400
```

To execute the code in the above cell, select it with a click and then either press the play button to the left of the code, or use the keyboard shortcut "Command/Ctrl+Enter". To edit the code, just click the cell and start editing.

Variables that you define in one cell can later be used in other cells:

```
seconds_in_a_week = 7 * seconds_in_a_day
seconds_in_a_week
```

```
604800
```

Colab notebooks allow you to combine **executable code** and **rich text** in a single document, along with **images**, **HTML**, **LaTeX** and more. When you create your own Colab notebooks, they are stored in your Google Drive account. You can easily share your Colab notebooks with co-workers or friends, allowing them to comment on your notebooks or even edit them. To learn more, see Overview of Colab. To create a new Colab notebook you can use the File menu above, or use the following link: create a new Colab notebook.

Colab notebooks are Jupyter notebooks that are hosted by Colab. To learn more about the Jupyter project, see jupyter.org.

Google Colab is available in VS Code!